Supplement S3: Observable Dataset

33 Relations Compared with Measurement; 62 Conditional and Diagnostic Entries

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S3.1 Introduction

This supplement documents the framework's observable dataset (`canonical/data/observables.json`, 95 entries) and the comparison data behind the main text. The dataset is a ledger, not a list of predictions. Version 3.6 counts **33 relations** (Type I) as the framework's claims, compared with measurement in main §4. The other **62 entries** (Types II–IV) are conditional extractions, outputs of mechanisms that consume calibration choices, or diagnostics of the approximate seam metric; they are kept here for traceability and are not counted (main §4.5).

Earlier versions described the dataset as "95 observables derived from a single G_2 metric, zero freely adjustable parameters". That description is withdrawn. The 33 relations are closed forms in twenty declared constants, assigned to observables by the author (main §3); no geometric map sends the invariants to the observables, and the metric does not produce them. The entries below carry a status that says which of these they are.

S3.2 Type classification

Type	Count	What it is	Counted in main §4
I	33	Dimensionless closed forms in the structural constants, assigned to measured parameters	yes
II	19	Dimensionful numbers read in a chosen unit (R10), scale bridges, one mixed-scale ratio (B16) and its cascade (B17), and legacy values without a documented derivation	no

Type	Count	What it is	Counted in main §4
III	21	Outputs of mechanisms on the approximate seam metric (Wilson-line splitting, associative volumes, spectral counts, bundle conditioning), with calibration choices; and the withdrawn supersymmetric running	no
IV	22	Structural quantities and numerical diagnostics, with no experimental counterpart	no

Type I. Each relation is a ratio of structural integers or uses one further operation ($\sqrt{}$, φ, ln 2, ζ values), declared in its status. The status vocabulary is that of main §4.1: *integer*, *algebraic / φ / ln 2 / ζ values*, *det(g)* (R2), κ_T (R3), *proximity*, *tension*, *approximate* (R8, R11, R12), *inside the PDG range*, *unit assignment* (R6); m_b/m_t is compared in the mass scheme declared in R7.

Type II. Several entries are numbers read in MeV or GeV, for example $m_u = \sqrt{}(\dim G_2/3) = 2.16$, $m_d = \ln 107 = 4.673$, $m_c = (\dim G_2 - \pi)^3 = 1280$, $m_t = 415^2$. Reading a structural number in a unit is an assignment of the same kind as reading 197 in degrees, with a far larger space of alternatives (R10). The CKM magnitudes (B10–B15) and the ratios m_b/m_d and m_t/m_c (B18, B19) are legacy values: their derivation is not documented in the current dataset, and they are retained for traceability, not counted as predictions. B18 and B19 are not fixed by the Type I quark ratios alone, which relate two separate groups, {u, d, s, c} and {b, t}. m_c/m_d (B17) combines B16, of Type II, with A08. Earlier versions described all nineteen as "a Type I ratio times an experimental anchor"; that description did not match the dataset and is withdrawn.

Type III. The Wilson-line and associative-volume entries reproduce lepton mass ratios at the percent level with declared calibration choices (a coupling $c = 0.452$ and a cycle assignment) selected against the measured ratios, and an exponent $\alpha = e^K = 0.002763$ computed from the Kähler potential of the approximate metric; they are conditional reconstructions (main §6.2). The comparison of C18 is mismatched: its value is $\mathrm{Vol}_e - \mathrm{Vol}_\mu = 3.271$, its reference is $\ln(m_\tau/m_\mu) = 2.823$, and the instanton relation pairs that reference with $\mathrm{Vol}_\mu - \mathrm{Vol}_\tau = 5.362$ (the matching reference for $\mathrm{Vol}_e - \mathrm{Vol}_\mu$ is $\ln(m_\mu/m_e) = 5.33$), so its 15.9% deviation carries no meaning. The cycle search scored this same pair. C20 uses the same difference and C21 is the ratio C19/C20; with $\mathrm{Vol}_\mu - \mathrm{Vol}_\tau$ in C20 they would read 16.79 and 207.6, still within 1%. The entries are kept as computed. The four entries of the two-loop supersymmetric running from a grand-unification boundary (C04–C07) are withdrawn with that reading (main §5). The remaining entries are diagnostics, some compared with a mathematical expectation (the Weyl exponent with d/2,

the gauge-kinetic conditioning with 1), none with a measurement.

Type IV. Topological inputs (b_2, b_3: R1), integer combinations (H*, N_{gen}), the imposed normalization det(g) = 65/32 (R2), numerical diagnostics of the certified seam metric (Newton–Kantorovich quantities, Gram conditioning, spectral counts, metric block eigenvalues), and three entries taken from the black-hole-remnant analysis of main ref. [42], one of which depends on the scale identification $\tau_0 = v_{EW}$.

S3.3 Experimental data and conventions

The comparison data were refreshed on 2026-10-22 to the most recent releases (`canonical/scripts/k7_v3_6_exp_refresh.py`, log `canonical/results/k7_v3_6_exp_refresh.json`):

Source	Used for	Convention
PDG 2026 [1]	particle masses, $\sin^2\theta_W$ and α_s at M_Z, CKM parameters, quark mass ratios	MS-bar at the PDG reference scales; m_t from direct measurements
NuFIT 6.1 [27]	neutrino mixing	normal ordering, with the tabulated Super-Kamiokande and IceCube atmospheric data; one variant for all four entries
Planck 2018 [3]	cosmology	base ΛCDM, TT,TE,EE+lowE+lensing; one column for all entries; the PR4 reanalysis [49] is not used
CODATA 2022 [50]	α at $q^2 = 0$, m_μ/m_e	n/a

Derived reference values. Mass ratios are computed from the published masses, with error propagation that neglects correlations. This is an approximation, not a bound: the summary tables do not publish the covariances, and for a ratio R = x/y the term -2 Cov(x, y)/(xy) in Var(R)/R^2 can have either sign. The density ratios A25, A26, A27, A29, A30 and A33 are computed sample by sample on the published Planck chains [3] (base_plikHM_TTTEEE_lowl_lowE_lensing, 25 225 weighted samples, whose marginal statistics reproduce the published table), as weighted mean and standard deviation; `canonical/scripts/k7_v3_6_planck_ratios.py`. No first-order propagation or proposal covariance enters. Earlier drafts propagated them without correlations, which gave $\Omega_c/\Omega_m = 0.839(11)$ at $+0.4\sigma$; ω_c and ω_m are correlated at about 0.99, and the chains give 0.8390(19), $+2.5\sigma$.

Total matter. Planck's Ω_m includes neutrinos ($\Omega_\nu h^2 = 0.00064$ at $\Sigma m_\nu = 0.06$ eV, $\Omega_\nu/\Omega_m = 0.0045$). So $\Omega_b/\Omega_m + \Omega_c/\Omega_m = 0.9955$ in the reference, while the formulas 5/32 and 27/32 sum to 1. This is a defect of compatibility with the Planck definition; the denominator is not replaced by $\Omega_b + \Omega_c$, since that would change the quantities tested. For information, that other quantity gives 0.8428(19) for $\Omega_c/(\Omega_b + \Omega_c)$, and pulls of $\pm 0.5\sigma$. The complementarity of baryons and cold dark matter, with neutrinos neglected,

does not require flatness. λ_H is the tree-level $m_H{}^2/(2v^2)$ with $v = (\sqrt{2}\, G_F)^{-1/2}$, a definition that is itself a scheme choice. m_s/m_d is given by the PDG as the range 17–22, without an uncertainty; it is stored as a range, and no pull is defined (the relation's value 20 lies inside it).

Confidence levels. All uncertainties are brought to 1σ. The PDG summary tables quote the light-quark masses, m_c, m_b and m_u/m_d at 90% confidence; these are divided by 1.645. The lattice average of m_c/m_s (PDG quark-mass review, statistical and systematic errors in quadrature) is already a 1σ value.

Asymmetric uncertainties. NuFIT and the CKM parameter A have asymmetric errors; both sides are kept in the dataset, and the pull uses the side facing the framework's value. The $\sin^2\theta_{23}$ profile is not Gaussian: its 3σ range (0.435–0.584) contains 6/11 although the Gaussian pull exceeds 4σ (main §4.3).

Corrections made in the refresh. The previous dataset gave m_μ/m_τ and m_τ/m_e with uncertainties larger than the PDG errors propagate (by factors 13 and 1.3); computed Q_{Koide} from the PDG 2022 τ mass while m_τ/m_e used the later one; gave m_b/m_t an uncertainty with no source; quoted an h that matched no Planck column; mixed NuFIT variants with and without atmospheric data; gave m_μ/m_e an uncertainty of 0.02 where CODATA 2022 gives 4.6×10^{-6}; and labelled the CKM entries A16 and A18 $\sin^2\theta$ although their values are $\sin\theta_{12} = \lambda$ and $\sin\theta_{23} = A\lambda^2$. With the corrected data and one convention for all uncertainties, the 33 relations fall into the five classes of main §4.2: 23 within 2σ, 3 in tension, 5 proximities beyond 3σ (m_b/m_t in the scheme of R7 among them), 1 inside the PDG range, 1 with a non-Gaussian profile. (The earlier count, 28 within 2σ, rested on the uncertainties listed above.)

Not refreshed. Y_p (0.245 ± 0.003) is the PDG's recommended observational helium abundance from HII regions (BBN review of the 2024 Review of Particle Physics [56], eq. 24.3; earlier versions of the dataset labelled it Planck 2018, which it is not), and was not re-checked against the 2026 edition; the Type II CKM magnitudes and mass cascades, and the entries C05, C17 and C18, keep their earlier reference values (PDG 2025), since they are not compared in the main text.

S3.4 Structural constants

The twenty constants, their origin and their status are given in main §3.1. Two corrections to earlier versions of this supplement: $N_{gen} = 3$ is an integer identity ($b_2/\dim K_7$), not the output of an index theorem (main §2.4); and b_2, b_3 are inputs posited by the framework (R1), not quantities confirmed by the spectral analysis of [B], which concerns an approximate metric.

S3.5 The 33 relations: reference data

Formulas, framework values and deviations are in main §4.1. This table gives, for each relation, the reference value, its source and the convention under which it is compared. It is generated from the dataset (`canonical/scripts/k7_v3_6_s3_tables.py`); the status column is read from main §4.1.

ID	Observable	Measured	Source	Pull	Convention / note	Status
A01	$\sin^2\theta_W$ (M_Z)	0.23122(6)	PDG 2026	-7.5σ	MS-bar at M_Z (PDG Table 1.1)	integer; **proximity**
A02	$\alpha_s(M_Z)$	0.1180(9)	PDG 2026	-0.2σ	n/a	algebraic
A03	λ_H	0.12914(23)	PDG 2026	-1.3σ	tree level ${m_H}^2/(2v^2)$, v = $(\sqrt{2}\,G_F)^{-1/2}$ = 246.220 GeV	algebraic; det(g)
A04	$\alpha^{-1}(0)$	137.035999177(21)	CODATA 2022	$\gg 10\sigma$	α^{-1} at $q^2 = 0$	det(g), κ_T; **proximity**
A05	Q_{Koide}	0.6666645(51)	PDG 2026	$+0.4\sigma$	computed from the PDG 2026 masses	integer; approximate (R8)
A06	m_τ/m_e	3477.37(18)	PDG 2026	-2.1σ	from the PDG 2026 masses	integer; approximate (R8); **tension**
A07	m_μ/m_e	206.7682827(46)	CODATA 2022	$\gg 10\sigma$	n/a	φ; approximate (R8); **proximity**
A08	m_s/m_d	17–22 (range)	PDG 2026	n/a	PDG gives the range 17–22, not a measurement with an uncertainty	integer; inside the PDG range
A09	m_c/m_s	11.769(35)	PDG 2026	-1.6σ	lattice average (N_f = 2+1+1), PDG quark-mass review	integer
A10	m_b/m_t	0.024253(43)	PDG 2026	-10.2σ	m_b at 90% CL converted to 1σ	integer; **proximity** (in the scheme declared in R7)
A11	m_u/m_d	0.462(12)	PDG 2026	$+0.7\sigma$	MS-bar at 2 GeV	integer
A12	$\sin^2\theta_{12}$ (PMNS)	0.3088(+67/−66)	NuFIT 6.1 (IC24+SK-atm, NO)	-0.2σ	n/a	integer
A13	$\sin^2\theta_{23}$ (PMNS)	0.470(+17/−14)	NuFIT 6.1 (IC24+SK-atm, NO)	$+4.4\sigma$*	3σ range 41.27°–49.86°, i.e. $\sin^2$ 0.435–0.584, contains 6/11	integer; **tension** (octant; inside the NuFIT 3σ range, main §4.3)
A14	$\sin^2\theta_{13}$ (PMNS)	0.02248(+55/−59)	NuFIT 6.1 (IC24+SK-atm, NO)	-0.5σ	n/a	integer
A15	δ_{CP}	212 (+26/−36)°	NuFIT 6.1 (IC24+SK-atm, NO)	-0.4σ	degrees	**unit assignment** (R6)
A16	sin θ_{12} (CKM)	0.22517(68)	PDG 2026	$+0.9\sigma$	PDG CKM review: sin $\theta_{12} = \lambda$ (not squared)	integer; approximate (R12)
A17	A (Wolfenstein)	0.826(+17/−15)	PDG 2026	$+0.7\sigma$	PDG CKM review, Wolfenstein fit	integer; approximate (R12)
A18	sin θ_{23} (CKM)	0.04189(+81/−69)	PDG 2026	-0.3σ	PDG CKM review: sin $\theta_{23} = A\lambda^2$ (not squared)	integer; approximate (R12)
A19	m_H/m_t	0.7250(13)	PDG 2026	$+1.8\sigma$	m_t from direct measurements	integer
A20	m_H/m_W	1.5571(14)	PDG 2026	$+0.5\sigma$	n/a	integer
A21	m_W/m_Z	0.88136(15)	PDG 2026	-2.7σ	W/Z mass ratio of the PDG summary table	integer; **tension**
A22	m_μ/m_τ	0.0594612(30)	PDG 2026	$+20.8\sigma$	from the PDG 2026 masses	integer; approximate (R8); **proximity**
A23	Ω_{DE}	0.6847(73)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+0.2\sigma$	Ω_Λ	ln 2; approximate (R11)

ID	Observable	Measured	Source	Pull	Convention / note	Status
A24	n_s	0.9649(42)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	n/a	ζ values
A25	Ω_{DM}/Ω_b	5.364(77)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+0.1\sigma$	$\Omega_c\ h^2$ / $\Omega_b\ h^2$	integer; approximate (R11)
A26	Ω_c/Ω_Λ	0.387(14)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	$\Omega_c\ h^2$ / h^2 / Ω_Λ	integer; det(g); approximate (R11)
A27	Ω_Λ/Ω_m	2.173(74)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	Ω_Λ / Ω_m, Planck definitions	integer; approximate (R11)
A28	h	0.6736(54)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	n/a	integer
A29	Ω_b/Ω_m	0.1565(19)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.1σ	$\Omega_b\ h^2$ / $\Omega_m\ h^2$, Ω_m includes neutrinos	integer; det(g); approximate (R11)
A30	Ω_c/Ω_m	0.8390(19)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+2.5\sigma$	$\Omega_c\ h^2$ / $\Omega_m\ h^2$, Ω_m includes neutrinos	integer; det(g); approximate (R11); **tension**
A31	σ_8	0.8111(60)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.3σ	n/a	integer; det(g)
A32	Y_p	0.245(3)	PDG 2024 (BBN review, eq. 24.3)	$+0.3\sigma$	n/a	integer; κ_T
A33	Ω_m/Ω_Λ	0.461(16)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.1σ	Ω_m / Ω_Λ, Planck definitions	integer; κ_T; approximate (R11)

* Non-Gaussian profile (S3.3).

S3.6 The 62 conditional and diagnostic entries

Generated from the dataset. "Reference value" is the measured value where one exists, otherwise the mathematical expectation the entry is compared with, otherwise n/a. None of these entries is counted as a prediction.

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
B01	II	m_u	sqrt(dim(G_2)/3)	2.16	2.16	unit assignment (R10)
B02	II	m_d	ln(107)	4.673	4.7	unit assignment (R10)
B03	II	m_s	$\tau\times24$	93.52	92.9	unit assignment (R10)
B04	II	m_c	(dim(G_2)-π)^3	1280	1272.9	unit assignment (R10)
B05	II	m_b	$2b_2\times$H*	4158	4186	unit assignment (R10)
B06	II	m_t	415^2	172225	172600	unit assignment (R10)
B07	II	v_{EW}	Scale bridge from the unit assignment (R10)	246.87	246.22	scale bridge; unit assignment (R10)
B08	II	M_W	$(g_2/2)\times v_{EW}$	80.4	80.3625	scale bridge; unit assignment (R10)
B09	II	M_Z	$M_W/\cos(\theta_W)$	91.2	91.1879	scale bridge; unit assignment (R10)
B10	II	$\|V_{us}\|$	not documented (legacy label: Cabibbo from b_2)	0.2245	0.2243	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B11	II	$\|V_{cb}\|$	not documented (legacy label: 2nd gen mixing)	0.04214	0.0422	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B12	II	$\|V_{ub}\|$	not documented (legacy label: 3rd gen coupling)	0.003947	0.00394	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B13	II	$\|V_{td}\|$	not documented (legacy label: down-type mixing)	0.008657	0.00867	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B14	II	$\|V_{ts}\|$	not documented (legacy label: strange-top coupling)	0.04154	0.0415	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B15	II	$\|V_{tb}\|$	not documented (legacy label: unitarity)	0.999106	0.999105	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B16	II	m_c/m_s (ext)	$\tau\times3.49$	13.6	13.6	mixed scales (m_c at m_c, m_s at 2 GeV); not comparable
B17	II	m_c/m_d	m_c/m_s (ext, B16) × m_s/m_d (A08) = $\tau\times3.49\times20$	271.99	272	cascade of B16 (Type II, not comparable) and A08; no independent content
B18	II	m_b/m_d	not documented (legacy label: cascade)	891.97	893	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
B19	II	m_t/m_c	not documented (legacy label: cascade)	135.49	136	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
C01	III	m_τ/m_e (dyn)	Wilson line 3-gen optimization	3403.1	3477.37	conditional reconstruction (calibration choices, main §6.2)
C02	III	m_τ/m_μ (dyn)	Wilson line 3-gen optimization	16.5441	16.8177	conditional reconstruction (calibration choices, main §6.2)
C03	III	c*	Non-adiabatic threshold	0.177828	n/a	diagnostic of the approximate seam metric
C04	III	$\sin^2\theta_W$ (RGE)	2-loop MSSM RGE	0.237655	0.23122	withdrawn (supersymmetric running from a GUT boundary, main §5)
C05	III	α_s (RGE)	2-loop MSSM RGE	0.103769	0.118	withdrawn (supersymmetric running from a GUT boundary, main §5)
C06	III	M_{GUT}	Gauge unification	2e+16	2e+16	withdrawn (supersymmetric running from a GUT boundary, main §5)
C07	III	${\alpha_{GUT}}^{-1}$	Gauge unification	25.3	25.3	withdrawn (supersymmetric running from a GUT boundary, main §5)
C08	III	λ_1	Neumann FD + Richardson	0.124607	n/a	diagnostic of the approximate seam metric
C09	III	α_{Weyl}	$N(\lambda) \sim C\ \lambda^{d/2}$	3.46002	3.5	diagnostic of the approximate seam metric (compared with d/2, not a measurement)
C10	III	N_{KK}	Unified KK tower enumeration	22671	n/a	diagnostic of the approximate seam metric
C11	III	N_{fiber}	Adiabatic + explicit fiber modes	57578	n/a	diagnostic of the approximate seam metric
C12	III	spacing	Level spacing statistics	1.35838	n/a	diagnostic of the approximate seam metric
C13	III	κ_{gauge}	f_{IJ} eigenvalue spread	1.0471	1	diagnostic of the approximate seam metric (compared with 1, not a measurement)
C14	III	n_{SD}	Count of positive eigenvalues of the K3 intersection form	3	3	diagnostic of the approximate seam metric (compared with N_{gen})
C15	III	N_{assoc}	Homology 3-cycles with φ-calibration	57	n/a	diagnostic of the approximate seam metric
C16	III	$\|\|T\|\|_C{}^0$	G_2 torsion norm	2.98e-05	n/a	diagnostic of the approximate seam metric
C17	III	ΔV(e-τ)	Associative volume differences	8.6331	8.15393	conditional reconstruction (calibration choices, main §6.2)
C18	III	ΔV(e-μ)	Associative volume differences	3.27125	2.82257	conditional reconstruction (calibration choices, main §6.2); mismatched reference (see Type III above)
C19	III	m_τ/m_e (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	3485.3	3477.37	conditional reconstruction (calibration choices, main §6.2)
C20	III	m_τ/m_μ (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	16.69	16.8177	conditional reconstruction (calibration choices, main §6.2); inherits the C18 pairing
C21	III	m_μ/m_e (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	208.8	206.768	conditional reconstruction (calibration choices, main §6.2); ratio C19/C20, inherits the C18 pairing
D01	IV	b_2	TCS K_7	21	n/a	input (R1)
D02	IV	b_3	TCS K_7	77	n/a	input (R1)
D03	IV	H*	b_2+b_3+1	99	n/a	integer combination of R1
D04	IV	N_{gen}	$b_2/\dim(K_7)$	3	n/a	integer identity (main §2.4)

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
D05	IV	det(g)	$\det(g_5)$	2.03125	n/a	imposed normalization (R2)
D06	IV	h_{NK}	$\beta\times\eta\times\gamma$	8.95e-09	n/a	diagnostic of the approximate seam metric
D07	IV	dist(g,g*)	NK contraction ball	4.86e-06	n/a	diagnostic of the approximate seam metric
D08	IV	$\|\|T\|\|_C{}^0$ (cert)	Chebyshev coefficient bound	2.95e-05	n/a	diagnostic of the approximate seam metric
D09	IV	$\kappa(G_{K3})$	L^2 Gram matrix K3 2-forms	1.05232	n/a	diagnostic of the approximate seam metric
D10	IV	N_{levels}	Unified spectrum enumeration	22671	n/a	diagnostic of the approximate seam metric
D11	IV	`N_states_total`	Including degeneracies	210060	n/a	diagnostic of the approximate seam metric
D12	IV	V_{eff}	Effective 7D volume from Weyl fit	5.38e+05	n/a	diagnostic of the approximate seam metric
D13	IV	$n_{homology}$	$b_3 = 77$	77	n/a	diagnostic of the approximate seam metric
D14	IV	N_{assoc}	φ-calibrated cycles	57	n/a	diagnostic of the approximate seam metric
D15	IV	$\|\|Y\|\|$	Total Yukawa coupling norm	8.96902	n/a	diagnostic of the approximate seam metric
D16	IV	ΔV_{range}	max(V) - min(V)	8.91952	n/a	diagnostic of the approximate seam metric
D17	IV	g_{ss}	$g_{ss} = 19/6$ (S^1 sector)	3.16667	n/a	diagnostic of the approximate seam metric
D18	IV	g_{T2}	$g_{T2} = 7/6$ (T^2 sector)	1.16667	n/a	diagnostic of the approximate seam metric
D19	IV	g_{K3}	$g_{K3} \sim 64/77$ (K3 sector mean)	0.831169	n/a	diagnostic of the approximate seam metric
D20	IV	N_{QNM}	b2 + b3 = H* - 1	98	n/a	from [42]; topological count
D21	IV	`b3/b3_S3S4`	b3(K7) = 77 vs b3(S3xS4) = 1	77	n/a	from [42]; topological count
D22	IV	M_{res}	$\text{tau}_0 = v_{EW}$, $M_{res} = v_{EW}{}^2 / M_{Pl}$	4969.2	n/a	from [42]; conditional on the scale identification $\tau_0 = v_{EW}$

Withdrawn readings attached to these entries. Earlier versions of this supplement listed "exact matches" (deviation below 0.01%) and ranked sectors by mean deviation. Both are withdrawn: an exact match between a Type II number and a PDG mass rounded to two or three figures ($m_u = 2.16$, $m_c = 1.27$ GeV) measures the rounding, not the framework; M_{GUT}, $\alpha_{GUT}{}^{-1}$ and rank(Y) were compared with values the same computation had set; and a mean deviation measures closeness, which main §7 shows is not distinctive. The description of Q_{Koide} as "the most precise agreement in the framework, with no fitting involved" is withdrawn with the primary-relation reading (main §6.1): $Q = 2/3$ predates the framework and every invented vocabulary reaches it. An earlier table of neutrino angles in this supplement used formulas ($\theta_{23} = \arcsin(75/99)$, θ_{12} from an auxiliary parameter) that are not the dataset's; the dataset's relations are those of main §4.1.

S3.7 Lean cross-references

The Lean formalization (Arithmon/K7-Lean, main §8) verifies arithmetic identities. A verified identity says that a closed form evaluates to a number; it says nothing about the physical identification, the choice

of formula, or the consistency of relations taken together. The table gives, for each of the 33 relations, the Lean declaration that ties the closed form to its value, and the scope of that proof. It is generated from `canonical/results/k7_v3_6_lean_map.json`; the library builds at the commit named, and the axioms are those reported by `#print axioms`.

Arithmon/K7-Lean, commit `34ce857` (merged into `main`, PR #167); aggregate `Certificate.Relations V36.relations_v36_certified`, which depends only on the axioms `propext`, `Classical.choice`, `Quot. sound`.

ID	Observable	Lean declaration (under `GIFT.`)	Location	Scope
A01	$\sin^2\theta_{\mathrm{W}}$	`Certificate.RelationsV36.A01_sin2_theta_W`	Certificate/ RelationsV36.lean:50	exact rational equality of the closed form
A02	$\alpha_{\mathrm{s}}(\mathrm{M_Z})$	`Certificate.RelationsV36.A02_alpha_s`	Certificate/ RelationsV36.lean:55	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A03	λ_{H}	`Certificate.RelationsV36.A03_lambda_H`	Certificate/ RelationsV36.lean:59	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A04	$\alpha^{-1}(0)$	`Certificate.RelationsV36.A04_alpha_inv`	Certificate/ RelationsV36.lean:65	exact rational equality of the closed form
A05	$\mathrm{Q_{Koide}}$	`Certificate.RelationsV36.A05_Q_Koide`	Certificate/ RelationsV36.lean:73	exact rational equality of the closed form
A06	$\mathrm{m_\tau/m_e}$	`Certificate.RelationsV36.A06_m_tau_m_e`	Certificate/ RelationsV36.lean:77	exact rational equality of the closed form
A07	$\mathrm{m_\mu/m_e}$	`Certificate.RelationsV36.A07_m_mu_m_e`	Certificate/ RelationsV36.lean:81	base 27 and interval bounds on 27^{φ}
A08	$\mathrm{m_s/m_d}$	`Certificate.RelationsV36.A08_m_s_m_d`	Certificate/ RelationsV36.lean:91	exact rational equality of the closed form
A09	$\mathrm{m_c/m_s}$	`Certificate.RelationsV36.A09_m_c_m_s`	Certificate/ RelationsV36.lean:95	exact rational equality of the closed form
A10	$\mathrm{m_b/m_t}$	`Certificate.RelationsV36.A10_m_b_m_t`	Certificate/ RelationsV36.lean:100	exact rational equality of the closed form
A11	$\mathrm{m_u/m_d}$	`Certificate.RelationsV36.A11_m_u_m_d`	Certificate/ RelationsV36.lean:104	exact rational equality of the closed form
A12	$\sin^2\theta_{12}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A12_sin2_theta12_PMNS`	Certificate/ RelationsV36.lean:109	exact rational equality of the closed form
A13	$\sin^2\theta_{23}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A13_sin2_theta23_PMNS`	Certificate/ RelationsV36.lean:113	exact rational equality of the closed form
A14	$\sin^2\theta_{13}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A14_sin2_theta13_PMNS`	Certificate/ RelationsV36.lean:117	exact rational equality of the closed form
A15	δ_{CP}	`Certificate.RelationsV36.A15_delta_CP`	Certificate/ RelationsV36.lean:122	the integer 197 (the unit is not in Lean)
A16	$\sin\theta_{12}{}^{\mathrm{CKM}}$	`Certificate.RelationsV36.A16_sin_theta12_CKM`	Certificate/ RelationsV36.lean:126	exact rational equality of the closed form
A17	$\mathrm{A_{Wolf}}$	`Certificate.RelationsV36.A17_A_Wolf`	Certificate/ RelationsV36.lean:131	exact rational equality of the closed form
A18	$\sin\theta_{23}{}^{\mathrm{CKM}}$	`Certificate.RelationsV36.A18_sin_theta23_CKM`	Certificate/ RelationsV36.lean:135	exact rational equality of the closed form
A19	$\mathrm{m_H/m_t}$	`Certificate.RelationsV36.A19_m_H_m_t`	Certificate/ RelationsV36.lean:140	exact rational equality of the closed form

ID	Observable	Lean declaration (under `GIFT.`)	Location	Scope
A20	m_H/m_W	`Certificate.RelationsV36.A20_m_H_m_W`	Certificate/ RelationsV36.lean:145	exact rational equality of the closed form
A21	m_W/m_Z	`Certificate.RelationsV36.A21_m_W_m_Z`	Certificate/ RelationsV36.lean:150	exact rational equality of the closed form
A22	m_μ/m_τ	`Certificate.RelationsV36.A22_m_mu_m_tau`	Certificate/ RelationsV36.lean:155	exact rational equality of the closed form
A23	Ω_{DE}	`Certificate.RelationsV36.A23_Omega_DE`	Certificate/ RelationsV36.lean:159	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A24	n_s	`Certificate.RelationsV36.A24_n_s`	Certificate/ RelationsV36.lean:165	indices (11, 5) and $\zeta(D_{bulk})/\zeta(\text{Weyl}) = \zeta(11)/\zeta(5)$ (no numerical bound)
A25	Ω_{DM}/Ω_b	`Certificate.RelationsV36.A25_Omega_DM_b`	Certificate/ RelationsV36.lean:173	exact rational equality of the closed form
A26	Ω_c/Ω_Λ	`Certificate.RelationsV36.A26_Omega_c_Lambda`	Certificate/ RelationsV36.lean:177	exact rational equality of the closed form
A27	Ω_Λ/Ω_m	`Certificate.RelationsV36.A27_Omega_Lambda_m`	Certificate/ RelationsV36.lean:182	exact rational equality of the closed form
A28	h	`Certificate.RelationsV36.A28_hubble_h`	Certificate/ RelationsV36.lean:187	exact rational equality of the closed form
A29	Ω_b/Ω_m	`Certificate.RelationsV36.A29_Omega_b_m`	Certificate/ RelationsV36.lean:191	exact rational equality of the closed form
A30	Ω_c/Ω_m	`Certificate.RelationsV36.A30_Omega_c_m`	Certificate/ RelationsV36.lean:196	exact rational equality of the closed form
A31	σ_8	`Certificate.RelationsV36.A31_sigma_8`	Certificate/ RelationsV36.lean:200	exact rational equality of the closed form
A32	Y_p	`Certificate.RelationsV36.A32_Y_p`	Certificate/ RelationsV36.lean:204	exact rational equality of the closed form
A33	Ω_m/Ω_Λ	`Certificate.RelationsV36.A33_Omega_m_Lambda`	Certificate/ RelationsV36.lean:208	exact rational equality of the closed form

Scope. Every relation has a statement that ties its closed form, written in the structural constants, to its value: an exact rational equality for 27; a closed form in $\mathbb{R}$ equal to the stated real value for α_s, λ_H and Ω_{DE}, with no numerical bound; the indices and the ζ-quotient for n_s, with no numerical bound; the base 27 and the bounds $206 < 27^\varphi < 209$ for m_μ/m_e, an interval wider than the deviation it is compared with; and the integer 197 for δ_{CP}, whose unit is a dictionary assignment outside Lean. The 33 statements are collected in `relations_v36_certified`.

What earlier versions cited. Earlier versions cited the single theorem `observables_certified` (`Certificate/Predictions.lean`) as certifying the relations. That theorem is a conjunction of statements of the form `X = v` where `X` is defined as the literal `v`; each holds by definition. Before the alignment of 2026-10-22 (commit `70dc589`), 24 relations had an exact statement elsewhere in the library, 5 had only their components, 1 had bounds and 3 (m_μ/m_τ, Ω_c/Ω_m, Ω_m/Ω_Λ) had none.

Names and withdrawn content. The alignment renamed declarations whose names misdescribed them, keeping the old names as deprecated aliases: the CKM entries are now `sin_theta12_CKM` and `sin_theta23_CKM` (the quantities compared are sin θ, S2.4); 42 is `two_b2 := 2 * b2`, not `chi_K7` ($\chi(K_7) =$

0, S1 §9.3); identities giving other values for the paper's observables (4/7 and 8/14 for θ_{23}, 7/23 for θ_{12}, 85/99) carry neutral arithmetic names and are documented as instances of the vocabulary freedom of S2.7. The δ_{CP} correction $197 \times 62/69 = 12214/69$ (S2.8) was removed from the aggregated certificate; its module is kept, marked withdrawn. A statement proves that the Koide parameter built from the framework's own $\mathrm{m}_\mu/\mathrm{m}_\mathrm{e} = 27^\varphi$ and $\mathrm{m}_\tau/\mathrm{m}_\mathrm{e} = 3477$ lies strictly below 2/3 (`Relations/KoideAssembly.lean`), the incompatibility of main §6.1.

Axioms. The formalization rests on fourteen axioms beyond Mathlib: four that import results from the literature (`cheeger_inequality`, `literature_package`, `spectral_upper_bound`, `neck_dominates`) and ten interval certificates that import numerical enclosures [ERR, §7.1]; no `sorry` is used.

S3.8 Data availability

The dataset is `canonical/data/observables.json` (95 entries; for each: formula, framework value, reference value and uncertainty, source, deviation, Lean theorem, provenance, and dated notes of every change). The refresh script and its log, and the generators of the main §4.1 table and of the S3 tables, are listed in S3.3 and S3.5. The dataset is deposited in the public repository (Arithmon/K7), as read by the look-elsewhere run in `publications/validation/v3.6/canonical/data/observables.json` and as a table in `publications/references/observables.csv`. The refresh script and the table generators are available from the author on request. The spectral companion artifacts of [B] are archived on Zenodo (DOI: 10.5281/zenodo.18920367).

Author's Related Works

- **[B]** B. de La Fournière, "Spectral Geometry of an Explicit G_2 Metric: Laplacian Spectrum and Harmonic Forms," Zenodo 10.5281/zenodo.18920367 (2026).
- **[ERR]** B. de La Fournière, "Erratum: GIFT / K_7 framework v3.5" (2026), repository Arithmon/K7, `publications/ERRATUM_v3.5.md`, commit aed460315eecc17a20402b1f4d20ac310dd4f6c2, github.com/Arithmon/K7/blob/aed4603/publications/ERRATUM_v3.5.md.

The K_7 Framework (formerly GIFT): Supplement S3: Observable Dataset 33 relations compared with measurement; 62 conditional and diagnostic entries

References

[1] Particle Data Group, F. Takahashi et al., Int. J. Mod. Phys. A 41, 2630011 (2026)

[2] S. Weinberg, Phys. Rev. D 13, 974 (1976)

[3] Planck Collaboration, A&A 641, A6 (2020); chains: Planck Legacy Archive, COM_CosmoParams_base-plikHM-TTTEEE-lowl-lowE_R3.00

[4] A.G. Riess et al., ApJL 934, L7 (2022)

[5] C.D. Froggatt, H.B. Nielsen, Nucl. Phys. B 147, 277 (1979)

[6] Y. Koide, Lett. Nuovo Cim. 34, 201 (1982)

[7] C. Furey, *Standard Model Physics from an Algebra?*, PhD thesis, U. Waterloo (2015)

[8] C. Furey, M.J. Hughes, Phys. Lett. B 831, 137186 (2022)

[9] R. Wilson, arXiv:2404.18938 (2024)

[10] T.P. Singh et al., arXiv:2206.06911v3 (2024)

[11] B.S. Acharya, S. Gukov, Phys. Rep. 392, 121 (2004)

[12] L. Foscolo et al., Duke Math. J. 170, 3 (2021)

[13] A. Kovalev, "Coassociative K3 fibrations of compact G_2-manifolds," arXiv:math/0511150 (2005)

[14] M. Haskins, J. Nordstrom, arXiv:1809.09083 (2022)

[15] G. Bera, "Associative submanifolds in twisted connected sum G_2-manifolds," arXiv:2209.00156 (2022)

[16] D. Crowley, S. Goette, J. Nordstrom, Invent. Math. (2025)

[17] T. Dray, C.A. Manogue, *The Geometry of the Octonions*, World Scientific (2015)

[18] J.F. Adams, *Lectures on Exceptional Lie Groups*, U. Chicago Press (1996)

[19] D.J. Gross et al., Nucl. Phys. B 256, 253 (1985)

[20] D.D. Joyce, *Compact Manifolds with Special Holonomy*, Oxford U. Press (2000)

[21] A. Kovalev, J. Reine Angew. Math. 565, 125 (2003)

[22] A. Corti et al., Duke Math. J. 164, 1971 (2015)

[23] R. Harvey, H.B. Lawson, Acta Math. 148, 47 (1982)

[24] B.S. Acharya, Class. Quant. Grav. 19, 5619 (2002)

[25] B.S. Acharya, E. Witten, arXiv:hep-th/0109152 (2001)

[26] The NOvA Collaboration, The T2K Collaboration, "Joint neutrino oscillation analysis from the T2K and NOvA experiments," Nature 646, 818-824 (2025), doi:10.1038/s41586-025-09599-3, arXiv:2510.19888

[27] I. Esteban, M.C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J.P. Pinheiro, T. Schwetz, "NuFit-6.0: updated global analysis of three-flavor neutrino oscillations," JHEP 12 (2024) 216, arXiv:2410.05380; NuFIT 6.1: three-neutrino fit based on data available in November 2025, www.nu-fit.org

[28] L. de Moura, S. Ullrich, CADE 28, 625 (2021)

[29] mathlib Community, github.com/leanprover-community/mathlib4

[30] DUNE Collaboration, FERMILAB-TM-2696 (2020)

[31] DUNE Collaboration, arXiv:2103.04797 (2021)

[32] E. Witten, "Strong coupling expansion of Calabi-Yau compactification," Nucl. Phys. B 471, 135 (1996)

[33] B.S. Acharya et al., Phys. Rev. D 76, 126010 (2007)

[34] M. Atiyah, E. Witten, Adv. Theor. Math. Phys. 6, 1 (2002)

[35] G. Kane, *String Theory and the Real World* (2017)

[36] J. Distler, S. Garibaldi, Commun. Math. Phys. 298, 419 (2010)

[37] J.C. Baez, "Octonions and the Standard Model," math.ucr.edu/home/baez/standard/ (2020-2025)

[38] Springer Nature, "Artificial intelligence (AI) policy," www.springernature.com/gp/policies (2024)

[39] J.A. Wheeler, "Information, physics, quantum: the search for links," in *Complexity, Entropy, and the Physics of Information* (W.H. Zurek, ed.), Addison-Wesley (1990), pp. 3–28.

[40] J. Worrall, "Structural realism: the best of both worlds?," *Dialectica* **43**, 99–124 (1989).

[41] J. Ladyman, D. Ross, *Every Thing Must Go: Metaphysics Naturalized*, Oxford University Press (2007).

[42] R. Pinčák, A. Pigazzini, M. Pudlák, E. Bartoš, "Geometric origin of a stable black hole remnant from torsion in G_2-manifold geometry," Gen. Rel. Grav. **58**, 29 (2026), doi:10.1007/s10714-026-03528-z

[43] D. Joyce, S. Karigiannis, "A new construction of compact torsion-free G_2-manifolds by gluing families of Eguchi-Hanson spaces," *J. Differential Geom.* (2021), arXiv:1707.09325 (2017).

[44] S. Mukai, "Finite groups of automorphisms of K3 surfaces and the Mathieu group," *Invent. Math.* **94**, 183–221 (1988).

[45] A. Garbagnati, A. Sarti, "Symplectic automorphisms of prime order on K3 surfaces," *J. Algebra* **318**, 323–350 (2007), arXiv:math/0603742.

[46] LEP2 SUSY Working Group (ALEPH, DELPHI, L3, OPAL), "Combined LEP chargino results, up to 208 GeV for large m0," note LEPSUSYWG/01-03.1 (2001), lepsusy.web.cern.ch.

[47] Fermilab News, "Fermilab marks major milestone for world-leading DUNE experiment," 7 May 2026, news.fnal.gov.

[48] J. Chen, H. Hong, "Intermediate curvature and splitting theorem," arXiv:2604.26529 (2026). (Cited in Supplement S1, §8.4.)

[49] M. Tristram et al., "Cosmological parameters derived from the final Planck data release (PR4)," A&A 682, A37 (2024). (Cited in Supplement S3; not used in the dataset.)

[50] E. Tiesinga, P.J. Mohr, D.B. Newell, B.N. Taylor, "CODATA recommended values of the fundamental physical constants: 2022," Rev. Mod. Phys. 97, 025002 (2025). (Cited in Supplement S3.)

[51] V. V. Nikulin, "Integer symmetric bilinear forms and some of their geometric applications," Izv. Akad. Nauk SSSR Ser. Mat. 43 (1979); English transl. Math. USSR Izv. 14 (1980).

[52] Belle II Collaboration, "Measurement of the τ-lepton mass with the Belle II experiment," Phys. Rev. D 108, 032006 (2023), doi:10.1103/PhysRevD.108.032006, arXiv:2305.19116.

[53] S. K. Donaldson, "Adiabatic limits of co-associative Kovalev–Lefschetz fibrations," in Algebra, Geometry, and Physics in the 21st Century, Progr. Math. 324, Birkhäuser (2017), pp. 1–29, arXiv:1603.08391.

[54] C. Camere, "Lattice polarized irreducible holomorphic symplectic manifolds," Ann. Inst. Fourier 66 (2016), 687–709, arXiv:1312.3224.

[55] C. Zhou, Z. Zhou, "Algebraic Stability and Cosmological Structure E: The Arithmetic Necessity of Three Generations of Fermions," Zenodo, doi:10.5281/zenodo.18626667 (2026).

[56] Particle Data Group, S. Navas et al., "Review of Particle Physics," Phys. Rev. D 110, 030001 (2024), doi:10.1103/PhysRevD.110.030001 (cited in S3 only)

[57] CDF Collaboration, T. Aaltonen et al., "High-precision measurement of the W boson mass with the CDF II detector," Science 376, 170 (2022), doi:10.1126/science.abk1781

[58] JUNO Collaboration, "First measurement of reactor neutrino oscillations at JUNO," arXiv:2511.14593 (2025).

[59] JUNO Collaboration, "Sub-percent precision measurement of neutrino oscillation parameters with JUNO," Chin. Phys. C 46, 123001 (2022), arXiv:2204.13249.

[60] D. Buttazzo, G. Degrassi, P.P. Giardino, G.F. Giudice, F. Sala, A. Salvio, A. Strumia, "Investigating the near-criticality of the Higgs boson," JHEP 12 (2013) 089, arXiv:1307.3536.